Syllabus Details

Syllabus ID:	14415
Syllabus Name:	**Data Handling and Visualization_Xử lý và Trực quan hóa Dữ liệu**
Course Name English:	**Data Handling & Visualization**
Subject Code:	**DHV301**
Learning-Teaching Method:	Inquiry-Based Learning, In-class lecture, Active Learning
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours (60 session) + 1h final exam + 90' practical exam + 102.5h self-study. EXTRA: 6 additional evaluation sessions for group project (Session 61–66) are outside the official 150h allocation
Pre-Requisite:	PDS301m
Description:	Upon completion of this course students should: 1. Understand the followings: • Data acquisition, data quality, and the end-to-end data handling pipeline • Data wrangling concepts: missing values, data types, normalization, encoding • Exploratory data analysis (EDA) and statistical reasoning (correlation, ANOVA, chi-square) • Visualization principles and data storytelling for different audiences • Dashboarding concepts and Power BI semantic modeling (star schema, relationships, measures) 2. Be able to: • Load data from CSV/Excel/JSON/SQL into Python (Pandas) and export results for reuse • Clean and transform data in Python and Excel/Power Query to prepare analytical datasets • Build effective charts with Matplotlib/Seaborn/Folium and communicate insights • Develop interactive Power BI reports/dashboards and publish to Power BI Service • Apply a simple predictive model (regression) to support data-driven decision making 3. Be able to work in team and present results with reproducible notebooks and dashboards
StudentTasks:	- Students must attend at least 80% of contact slots in order to be accepted to the final examination. - Student is responsible to do all exercises, assignments and labs given by instructor in class or at home and submit on time - Use laptop in class only for learning purpose - Promptly access to the FU FLM at https://flm.fpt.edu.vn/ for up-to-date course information
Tools:	- Internet - Laptop - Python 3.x, Jupyter Notebook (Anaconda) or VS Code - Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, folium - Microsoft Excel (Microsoft 365) and Power BI Desktop - Optional: SQLite (sqlite3), DAX Studio (for advanced DAX debugging) - LMS/FLM for submissions and resources
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	968/QĐ-ĐHFPT dated 08/07/2026
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	8/7/2026

7 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter	Wes McKinney	O'Reilly Media	2022	3rd	978-1098104030	True	False	True	
2	Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures	Claus O. Wilke	O'Reilly Media	2019	1st	978-1492031086	False	False	True	
3	Microsoft Power BI Cookbook: Convert raw data into business insights with updated techniques, use cases, and best practices	Greg Deckler, Brett Powell	Packt	2023	3rd	978-1835464274	False	True	False	
4	pandas User Guide: IO tools	pandas developers	pandas.pydata.org				False	False	True	Docs (Pandas): https://pandas.pyda IO guide: https://pandas.pydata.org/docs/use Recommended cloud practice platfo Colab (https://colab.research.google Kaggle Notebooks (https://www.kaggle.com/code).
5	Matplotlib Pyplot tutorial	Matplotlib developers	matplotlib.org				False	False	True	Docs (Matplotlib/Seaborn): https://matplotlib.org/stable/index.h https://matplotlib.org/stable/tutorials https://seaborn.pydata.org/
6	Power BI documentation	Microsoft	learn.microsoft.com				False	False	True	Docs (Power BI & DAX): https://learn.microsoft.com/en-us/p https://learn.microsoft.com/en-us/d
7	Power BI guidance: Star schema	Microsoft	learn.microsoft.com				False	False	True	Power BI data modeling (Star schem https://learn.microsoft.com/en-us/p bi/guidance/star-schema

4 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Apply data handling techniques (import, clean, transform, summarize) to prepare datasets for analysis in Python/Excel/Power Query.
2	CLO2	Communicate insights using data storytelling: select audience-appropriate narratives, annotate visuals, and write concise interpretations.
3	CLO3	Create clear and accurate visualizations in Python (Matplotlib/Seaborn/Folium), choosing suitable chart types and design best practices.
4	CLO4	Build interactive dashboards in Power BI using good data modeling (relationships/star schema) and DAX measures, then publish and share responsibly.

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Module 1 – Data Foundations & Sourcing M1.1 Problem framing & dataset documentation (Part 1/3) • Domain questions, stakeholders, KPIs • Data dictionary, schema, data types • Concepts & instructor demo	Offline	CLO1	IT	Core: Python for Data Analysis (McKinney) Ch. 1–2 (overview, Python, pandas basics). Prep: Install Anaconda/Miniconda + VS Code/Jupyter; download starter dataset & notebook. Refs: pandas IO guide (CSV/Excel); data documentation template (data dictionary). Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	DHV301	Before class: skim materials & identify dataset/business questions. In class: follow demo; take notes. After class: complete short warm-up exercises.	
2	Module 1 – Data Foundations & Sourcing M1.1 Problem framing & dataset documentation (Part 2/3) • Tooling setup (Python/Excel/Power BI) • Guided practice (hands-on)	Offline	CLO1	IT	Core: Python for Data Analysis (McKinney) Ch. 1–2 (overview, Python, pandas basics). Prep: Install Anaconda/Miniconda + VS Code/Jupyter; download starter dataset & notebook. Refs: pandas IO guide (CSV/Excel); data documentation template (data dictionary). Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link2-1 link2-2 link2-3	In class: complete guided lab steps; record outputs. After class: finish practice tasks; update notebook/report draft.	
3	Module 1 – Data Foundations & Sourcing M1.1 Problem framing & dataset documentation (Part 3/3) • Tooling setup (Python/Excel/Power BI) • Lab + checkpoint / reflection	Offline	CLO1	IT	Core: Python for Data Analysis (McKinney) Ch. 1–2 (overview, Python, pandas basics). Prep: Install Anaconda/Miniconda + VS Code/Jupyter; download starter dataset & notebook. Refs: pandas IO guide (CSV/Excel); data documentation template (data dictionary). Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link3-1 link3-2 link3-3	In class: finish mini-lab and checkpoint. After class: submit checkpoint deliverables (code + brief answers).	
4	Module 1 – Data Foundations & Sourcing M1.2 Data acquisition & I/O workflows (Part 1/3) • CSV/Excel/JSON basics; encoding; delimiters • Intro SQL (SQLite) for data access • Concepts & instructor demo	Offline	CLO1	IT	Core: McKinney Ch. 6 (Data Loading, Storage, File Formats) selected; pandas read_csv/read_excel docs. Optional: SQLite intro (SELECT, WHERE, GROUP BY) cheatsheet. Assets: multi-source dataset pack (CSV + Excel + JSON). Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link4-1 link4-2 link4-3 link4-4	Before class: skim materials & identify dataset/business questions. In class: follow demo; take notes. After class: complete short warm-up exercises.	
5	Module 1 – Data Foundations & Sourcing M1.2 Data acquisition & I/O workflows (Part 2/3) • Export formats & versioning datasets • Guided practice (hands-on)	Offline	CLO1	IT	Core: McKinney Ch. 6 (Data Loading, Storage, File Formats) selected; pandas read_csv/read_excel docs. Optional: SQLite intro (SELECT, WHERE, GROUP BY) cheatsheet. Assets: multi-source dataset pack (CSV + Excel + JSON). Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link5-1 link5-2 link5-3 link5-4	In class: complete guided lab steps; record outputs. After class: finish practice tasks; update notebook/report draft.	

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6	Module 1 – Data Foundations & Sourcing M1.2 Data acquisition & I/O workflows (Part 3/3) • Export formats & versioning datasets • Lab + checkpoint / reflection	Offline	CLO1	IT	Core: McKinney Ch. 6 (Data Loading, Storage, File Formats) selected; pandas read_csv/read_excel docs. Optional: SQLite intro (SELECT, WHERE, GROUP BY) cheatsheet. Assets: multi-source dataset pack (CSV + Excel + JSON). Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link6-1 link6-2 link6-3 link6-4	In class: finish mini-lab and checkpoint. After class: submit checkpoint deliverables (code + brief answers).	
7	Module 2 – Data Cleaning & Preparation M2.1 Quality issues & missing data (Part 1/3) • Missingness patterns; drop vs impute • Type casting; parsing dates; string cleanup • Concepts & instructor demo	Offline	CLO1	IT	Core: McKinney Ch. 7 (Data Cleaning & Preparation) – missing data. Refs: Coursera 'Preparing Data for Analysis (Excel)' – missing/dirty data patterns. Assets: messy dataset (missing, inconsistent dates/strings). Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link7-1 link7-2 link7-3	Before class: skim materials & identify dataset/business questions. In class: follow demo; take notes. After class: complete short warm-up exercises.	
8	Module 2 – Data Cleaning & Preparation M2.1 Quality issues & missing data (Part 2/3) • Reproducible cleaning log • Guided practice (hands-on)	Offline	CLO1	IT	Core: McKinney Ch. 7 (Data Cleaning & Preparation) – missing data. Refs: Coursera 'Preparing Data for Analysis (Excel)' – missing/dirty data patterns. Assets: messy dataset (missing, inconsistent dates/strings). Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link8-1 link8-2 link8-3	In class: complete guided lab steps; record outputs. After class: finish practice tasks; update notebook/report draft.	
9	Module 2 – Data Cleaning & Preparation M2.1 Quality issues & missing data (Part 3/3) • Reproducible cleaning log • Lab + checkpoint / reflection	Offline	CLO1	IT	Core: McKinney Ch. 7 (Data Cleaning & Preparation) – missing data. Refs: Coursera 'Preparing Data for Analysis (Excel)' – missing/dirty data patterns. Assets: messy dataset (missing, inconsistent dates/strings).	link9-1	In class: finish mini-lab and checkpoint. After class: submit checkpoint deliverables (code + brief answers).	
10	Module 2 – Data Cleaning & Preparation M2.2 Transformation & feature preparation (Part 1/3) • Normalization/scaling; binning • Categorical encoding (one-hot) • Concepts & instructor demo	Offline	CLO1	IT	Core: McKinney Ch. 7 – transform, replace, map, cut; pandas get_dummies. Refs: scikit-learn preprocessing overview (scaling/encoding) (short). Assets: categorical-heavy dataset for encoding.	link10-1	Before class: skim materials & identify dataset/business questions. In class: follow demo; take notes. After class: complete short warm-up exercises.	
11	Module 2 – Data Cleaning & Preparation M2.2 Transformation & feature preparation (Part 2/3) • Intro feature engineering + leakage awareness • Guided practice (hands-on)	Offline	CLO1	IT	Core: McKinney Ch. 7 – transform, replace, map, cut; pandas get_dummies. Refs: scikit-learn preprocessing overview (scaling/encoding) (short). Assets: categorical-heavy dataset for encoding.	link11-1	In class: complete guided lab steps; record outputs. After class: finish practice tasks; update notebook/report draft.	

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12	Module 2 – Data Cleaning & Preparation M2.2 Transformation & feature preparation (Part 3/3) • Intro feature engineering + leakage awareness • Lab + checkpoint / reflection	Offline	CLO1	IT	Core: McKinney Ch. 7 – transform, replace, map, cut; pandas get_dummies. Refs: scikit-learn preprocessing overview (scaling/encoding) (short). Assets: categorical-heavy dataset for encoding.	link12-1	In class: finish mini-lab and checkpoint. After class: submit checkpoint deliverables (code + brief answers).	
13	Module 3 – EDA & Statistical Reasoning M3.1 Descriptive statistics & distributions (Part 1/3) • Summary tables; percentiles; outliers • Visual EDA: hist/box • Concepts & instructor demo	Offline	CLO1, CLO3	IT	Core: McKinney Ch. 9 (Data Aggregation & Group Operations) + descriptive stats. Refs: Matplotlib intro (hist/box) + chart selection rules handout. Assets: EDA notebook starter.	link13-1	Before class: skim materials & identify dataset/business questions. In class: follow demo; take notes. After class: complete short warm-up exercises.	
14	Module 3 – EDA & Statistical Reasoning M3.1 Descriptive statistics & distributions (Part 2/3) • Interpreting skew, spread, anomalies • Guided practice (hands-on)	Offline	CLO1, CLO3	IT	Core: McKinney Ch. 9 (Data Aggregation & Group Operations) + descriptive stats. Refs: Matplotlib intro (hist/box) + chart selection rules handout. Assets: EDA notebook starter.	link14-1	In class: complete guided lab steps; record outputs. After class: finish practice tasks; update notebook/report draft.	
15	Module 3 – EDA & Statistical Reasoning M3.1 Descriptive statistics & distributions (Part 3/3) • Interpreting skew, spread, anomalies • Lab + checkpoint / reflection	Offline	CLO1, CLO3	IT	Core: McKinney Ch. 9 (Data Aggregation & Group Operations) + descriptive stats. Refs: Matplotlib intro (hist/box) + chart selection rules handout. Assets: EDA notebook starter.	link15-1	In class: finish mini-lab and checkpoint. After class: submit checkpoint deliverables (code + brief answers).	
16	Module 3 – EDA & Statistical Reasoning M3.2 Relationships & group comparisons (Part 1/3) • Groupby/aggregation • Correlation (Pearson/Spearman) & pitfalls • Concepts & instructor demo	Offline	CLO1, CLO3	IT	Core: aggregation/groupby/pivot; correlation caveats. Refs: Harvard CS109A materials on correlation & inference (selected). Assets: grouped dataset (segments, categories).	link16-1	Before class: skim materials & identify dataset/business questions. In class: follow demo; take notes. After class: complete short warm-up exercises.	
17	Module 3 – EDA & Statistical Reasoning M3.2 Relationships & group comparisons (Part 2/3) • Hypothesis tests: chi-square / ANOVA (use + interpret) • Guided practice (hands-on)	Offline	CLO1, CLO3	IT	Core: aggregation/groupby/pivot; correlation caveats. Refs: Harvard CS109A materials on correlation & inference (selected). Assets: grouped dataset (segments, categories).	link17-1	In class: complete guided lab steps; record outputs. After class: finish practice tasks; update notebook/report draft.	

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18	Module 3 – EDA & Statistical Reasoning M3.2 Relationships & group comparisons (Part 3/3) • Hypothesis tests: chi-square / ANOVA (use + interpret) • Lab + checkpoint / reflection	Offline	CLO1, CLO3	IT	Core: aggregation/groupby/pivot; correlation caveats. Refs: Harvard CS109A materials on correlation & inference (selected). Assets: grouped dataset (segments, categories).	link18-1	In class: finish mini-lab and checkpoint. After class: submit checkpoint deliverables (code + brief answers).	
19	Module 3 – EDA & Statistical Reasoning M3.3 Regression workflow (baseline) (Part 1/3) • Train/test split; metrics (MAE/RMSE/R^2) • Baseline vs improved models • Concepts & instructor demo	Offline	CLO1	IT	Core: scikit-learn linear regression workflow; train/test split; evaluation (MAE/RMSE/R2). Assets: regression lab notebook + dataset (house prices).	link19-1	Before class: skim materials & identify dataset/business questions. In class: follow demo; take notes. After class: complete short warm-up exercises.	
20	Module 3 – EDA & Statistical Reasoning M3.3 Regression workflow (baseline) (Part 2/3) • Communicating model results for decisions • Guided practice (hands-on)	Offline	CLO1	IT	Core: scikit-learn linear regression workflow; train/test split; evaluation (MAE/RMSE/R2). Assets: regression lab notebook + dataset (house prices).	link20-1	In class: complete guided lab steps; record outputs. After class: finish practice tasks; update notebook/report draft.	
21	Module 3 – EDA & Statistical Reasoning M3.3 Regression workflow (baseline) (Part 3/3) • Communicating model results for decisions • Lab + checkpoint / reflection	Offline	CLO1	IT	Core: scikit-learn linear regression workflow; train/test split; evaluation (MAE/RMSE/R2). Assets: regression lab notebook + dataset (house prices).	link21-1	In class: finish mini-lab and checkpoint. After class: submit checkpoint deliverables (code + brief answers).	
22	Module 3 – EDA & Statistical Reasoning M3.4 Refinement & pipelines (Part 1/3) • Polynomial features; regularization (Ridge) • Pipelines + preprocessing • Concepts & instructor demo	Offline	CLO1	IT	Core: model refinement & comparison; cross-validation concept (light). Optional: regularization intuition (Ridge/Lasso). Assets: ablation worksheet (what changed, why).	link22-1	Before class: skim materials & identify dataset/business questions. In class: follow demo; take notes. After class: complete short warm-up exercises.	
23	Module 3 – EDA & Statistical Reasoning M3.4 Refinement & pipelines (Part 2/3) • Simple tuning (GridSearchCV) + validation hygiene • Guided practice (hands-on)	Offline	CLO1	IT	Core: model refinement & comparison; cross-validation concept (light). Optional: regularization intuition (Ridge/Lasso). Assets: ablation worksheet (what changed, why).	link23-1	In class: complete guided lab steps; record outputs. After class: finish practice tasks; update notebook/report draft.	

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24	Module 3 – EDA & Statistical Reasoning M3.4 Refinement & pipelines (Part 3/3) • Simple tuning (GridSearchCV) + validation hygiene • Lab + checkpoint / reflection	Offline	CLO1	IT	Core: model refinement & comparison; cross-validation concept (light). Optional: regularization intuition (Ridge/Lasso). Assets: ablation worksheet (what changed, why).	link24-1	In class: finish mini-lab and checkpoint. After class: submit checkpoint deliverables (code + brief answers).	
25	Module 4 – Storytelling & Visualization in Python M4.1 Data storytelling fundamentals (Part 1/3) • Audience & message; narrative structure • Annotation & chart titles that “say the insight” • Concepts & instructor demo	Offline	CLO2	IT	Core: Data storytelling framework (question → evidence → insight → action). Refs: Fundamentals of Data Visualization (Wilke) – principles; example narratives. Assets: storytelling canvas template. Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link25-1 link25-2 link25-3	Before class: skim materials & identify dataset/business questions. In class: follow demo; take notes. After class: complete short warm-up exercises.	
26	Module 4 – Storytelling & Visualization in Python M4.1 Data storytelling fundamentals (Part 2/3) • Common chart mistakes & fixes • Guided practice (hands-on)	Offline	CLO2	IT	Core: Data storytelling framework (question → evidence → insight → action). Refs: Fundamentals of Data Visualization (Wilke) – principles; example narratives. Assets: storytelling canvas template. Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link26-1 link26-2 link26-3	In class: complete guided lab steps; record outputs. After class: finish practice tasks; update notebook/report draft.	
27	Module 4 – Storytelling & Visualization in Python M4.1 Data storytelling fundamentals (Part 3/3) • Common chart mistakes & fixes • Lab + checkpoint / reflection	Offline	CLO2	IT	Core: Data storytelling framework (question → evidence → insight → action). Refs: Fundamentals of Data Visualization (Wilke) – principles; example narratives. Assets: storytelling canvas template. Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link27-1 link27-2 link27-3	In class: finish mini-lab and checkpoint. After class: submit checkpoint deliverables (code + brief answers).	
28	Module 4 – Storytelling & Visualization in Python M4.2 Matplotlib foundations (Part 1/3) • Figure/axes, labels, scales, ticks • Line charts + time series basics • Concepts & instructor demo	Offline	CLO3	IT	Core: Matplotlib basics (figure/axes), labels, scales, styles. Refs: Matplotlib pyplot tutorial; Wilke – readability. Assets: plotting starter notebook. Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link28-1 link28-2 link28-3	Before class: skim materials & identify dataset/business questions. In class: follow demo; take notes. After class: complete short warm-up exercises.	
29	Module 4 – Storytelling & Visualization in Python M4.2 Matplotlib foundations (Part 2/3) • Saving figures & reproducibility • Guided practice (hands-on)	Offline	CLO3	IT	Core: Matplotlib basics (figure/axes), labels, scales, styles. Refs: Matplotlib pyplot tutorial; Wilke – readability. Assets: plotting starter notebook. Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link29-1 link29-2 link29-3	In class: complete guided lab steps; record outputs. After class: finish practice tasks; update notebook/report draft.	

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30	Module 4 – Storytelling & Visualization in Python M4.2 Matplotlib foundations (Part 3/3) • Saving figures & reproducibility • Lab + checkpoint / reflection	Offline	CLO3	IT	Core: Matplotlib basics (figure/axes), labels, scales, styles. Refs: Matplotlib pyplot tutorial; Wilke – readability. Assets: plotting starter notebook. Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link30-1 link30-2 link30-3	In class: finish mini-lab and checkpoint. After class: submit checkpoint deliverables (code + brief answers).	
31	Module 4 – Storytelling & Visualization in Python M4.3 Univariate & categorical views (Part 1/3) • Histograms, bars, area charts • Choosing bins; ordering categories • Concepts & instructor demo	Offline	CLO3	IT	Core: Seaborn statistical plots; categorical plots; faceting. Refs: Seaborn docs (tutorial/relational/categorical). Assets: dataset with categories & time. Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link31-1 link31-2 link31-3	Before class: skim materials & identify dataset/business questions. In class: follow demo; take notes. After class: complete short warm-up exercises.	
32	Module 4 – Storytelling & Visualization in Python M4.3 Univariate & categorical views (Part 2/3) • Styling for clarity (color used sparingly) • Guided practice (hands-on)	Offline	CLO3	IT	Core: Seaborn statistical plots; categorical plots; faceting. Refs: Seaborn docs (tutorial/relational/categorical). Assets: dataset with categories & time. Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link32-1 link32-2 link32-3	In class: complete guided lab steps; record outputs. After class: finish practice tasks; update notebook/report draft.	
33	Module 4 – Storytelling & Visualization in Python M4.3 Univariate & categorical views (Part 3/3) • Styling for clarity (color used sparingly) • Lab + checkpoint / reflection	Offline	CLO3	IT	Core: Seaborn statistical plots; categorical plots; faceting. Refs: Seaborn docs (tutorial/relational/categorical). Assets: dataset with categories & time. Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link33-1 link33-2 link33-3	In class: finish mini-lab and checkpoint. After class: submit checkpoint deliverables (code + brief answers).	
34	Module 4 – Storytelling & Visualization in Python M4.4 Bivariate & multivariate views (Part 1/3) • Scatter/bubble; box plots; faceting concept • Trend lines; overplotting fixes • Concepts & instructor demo	Offline	CLO3	IT	Core: multivariate plots; heatmap; pairplot; annotation best practices. Assets: correlation/heatmap lab notebook. Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link34-1 link34-2	Before class: skim materials & identify dataset/business questions. In class: follow demo; take notes. After class: complete short warm-up exercises.	
35	Module 4 – Storytelling & Visualization in Python M4.4 Bivariate & multivariate views (Part 2/3) • Reading relationships vs causation warnings • Guided practice (hands-on)	Offline	CLO3	IT	Core: multivariate plots; heatmap; pairplot; annotation best practices. Assets: correlation/heatmap lab notebook. Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link35-1 link35-2	In class: complete guided lab steps; record outputs. After class: finish practice tasks; update notebook/report draft.	

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36	Module 4 – Storytelling & Visualization in Python M4.4 Bivariate & multivariate views (Part 3/3) • Reading relationships vs causation warnings • Lab + checkpoint / reflection	Offline	CLO3	IT	Core: multivariate plots; heatmap; pairplot; annotation best practices. Assets: correlation/heatmap lab notebook. Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link36-1 link36-2	In class: finish mini-lab and checkpoint. After class: submit checkpoint deliverables (code + brief answers).	
37	Module 4 – Storytelling & Visualization in Python M4.5 Advanced charts (selective) (Part 1/3) • Waffle / wordcloud (when appropriate) • Seaborn regression plots • Concepts & instructor demo	Offline	CLO3, CLO2	IT	Optional/advanced chart types (only when suitable): small multiples, slope, waffle/wordcloud (careful). Core reading: Wilke chapter on chart types & misuse. Assets: chart gallery handout. Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link37-1 link37-2 link37-3	Before class: skim materials & identify dataset/business questions. In class: follow demo; take notes. After class: complete short warm-up exercises.	
38	Module 4 – Storytelling & Visualization in Python M4.5 Advanced charts (selective) (Part 2/3) • Design critique & improvement checklist • Guided practice (hands-on)	Offline	CLO3, CLO2	IT	Optional/advanced chart types (only when suitable): small multiples, slope, waffle/wordcloud (careful). Core reading: Wilke chapter on chart types & misuse. Assets: chart gallery handout. Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link38-1 link38-2 link38-3	In class: complete guided lab steps; record outputs. After class: finish practice tasks; update notebook/report draft.	
39	Module 4 – Storytelling & Visualization in Python M4.5 Advanced charts (selective) (Part 3/3) • Design critique & improvement checklist • Lab + checkpoint / reflection	Offline	CLO3, CLO2	IT	Optional/advanced chart types (only when suitable): small multiples, slope, waffle/wordcloud (careful). Core reading: Wilke chapter on chart types & misuse. Assets: chart gallery handout. Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link39-1 link39-2 link39-3	In class: finish mini-lab and checkpoint. After class: submit checkpoint deliverables (code + brief answers).	
40	Module 4 – Storytelling & Visualization in Python M4.6 Geospatial visualization (Folium) (Part 1/3) • Markers, layers, popups • GeoJSON + choropleth • Concepts & instructor demo	Offline	CLO3	IT	Core: geospatial visualization with Folium (markers, choropleth). Refs: Folium choropleth guide; data ethics note on maps. Assets: GeoJSON sample + location dataset. Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link40-1 link40-2 link40-3	Before class: skim materials & identify dataset/business questions. In class: follow demo; take notes. After class: complete short warm-up exercises.	
41	Module 4 – Storytelling & Visualization in Python M4.6 Geospatial visualization (Folium) (Part 2/3) • Map ethics: normalization & misleading scales • Guided practice (hands-on)	Offline	CLO3	IT	Core: geospatial visualization with Folium (markers, choropleth). Refs: Folium choropleth guide; data ethics note on maps. Assets: GeoJSON sample + location dataset. Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link41-1 link41-2 link41-3	In class: complete guided lab steps; record outputs. After class: finish practice tasks; update notebook/report draft.	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
42	Module 4 – Storytelling & Visualization in Python M4.6 Geospatial visualization (Folium) (Part 3/3) • Map ethics: normalization & misleading scales • Lab + checkpoint / reflection	Offline	CLO3	IT	Core: geospatial visualization with Folium (markers, choropleth). Refs: Folium choropleth guide; data ethics note on maps. Assets: GeoJSON sample + location dataset. Practice environment: local Jupyter, Google Colab, or Kaggle Notebooks may be used for Python-based work.	link42-1 link42-2 link42-3	In class: finish mini-lab and checkpoint. After class: submit checkpoint deliverables (code + brief answers).	
43	Module 5 – BI Dashboarding with Power BI M5.1 Power Query + data model (Part 1/3) • ETL steps; data types; merge/append • Star schema; relationships & granularity • Concepts & instructor demo	Offline	CLO4	IT	Core: Power BI Desktop – Power Query (ETL) + data model basics (star schema). Refs: Microsoft Learn: Power Query overview; modeling guidance. Assets: PBIX starter + dataset.	link43-1 link43-2	Before class: skim materials & identify dataset/business questions. In class: follow demo; take notes. After class: complete short warm-up exercises.	
44	Module 5 – BI Dashboarding with Power BI M5.1 Power Query + data model (Part 2/3) • Model documentation • Guided practice (hands-on)	Offline	CLO4	IT	Core: Power BI Desktop – Power Query (ETL) + data model basics (star schema). Refs: Microsoft Learn: Power Query overview; modeling guidance. Assets: PBIX starter + dataset.	link44-1 link44-2	In class: complete guided lab steps; record outputs. After class: finish practice tasks; update notebook/report draft.	
45	Module 5 – BI Dashboarding with Power BI M5.1 Power Query + data model (Part 3/3) • Model documentation • Lab + checkpoint / reflection	Offline	CLO4	IT	Core: Power BI Desktop – Power Query (ETL) + data model basics (star schema). Refs: Microsoft Learn: Power Query overview; modeling guidance. Assets: PBIX starter + dataset.	link45-1 link45-2	In class: finish mini-lab and checkpoint. After class: submit checkpoint deliverables (code + brief answers).	
46	Module 5 – BI Dashboarding with Power BI M5.2 DAX measures + publishing (Part 1/3) • Measures vs calculated columns • Core DAX patterns (CALCULATE, FILTER context) • Concepts & instructor demo	Offline	CLO4	IT	Core: DAX basics (measures, CALCULATE), visuals, slicers, drill-through; publish basics. Refs: Microsoft Learn DAX fundamentals. Assets: DAX practice sheet.	link46-1 link46-2	Before class: skim materials & identify dataset/business questions. In class: follow demo; take notes. After class: complete short warm-up exercises.	
47	Module 5 – BI Dashboarding with Power BI M5.2 DAX measures + publishing (Part 2/3) • Report pages, interactivity, publish & share • Guided practice (hands-on)	Offline	CLO4	IT	Core: DAX basics (measures, CALCULATE), visuals, slicers, drill-through; publish basics. Refs: Microsoft Learn DAX fundamentals. Assets: DAX practice sheet.	link47-1 link47-2	In class: complete guided lab steps; record outputs. After class: finish practice tasks; update notebook/report draft.	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
48	Module 5 – BI Dashboarding with Power BI M5.2 DAX measures + publishing (Part 3/3) • Report pages, interactivity, publish & share • Lab + checkpoint / reflection	Offline	CLO4	IT	Core: DAX basics (measures, CALCULATE), visuals, slicers, drill-through; publish basics. Refs: Microsoft Learn DAX fundamentals. Assets: DAX practice sheet.	link48-1 link48-2	In class: finish mini-lab and checkpoint. After class: submit checkpoint deliverables (code + brief answers).	
49	Module 6 – Responsible AI & Integrated Analytics M6.1 GenAI-assisted workflow (practical) (Part 1/3) • Use cases: chart ideation, code scaffolding, narrative drafts • Verification checklist (no hallucinations) • Concepts & instructor demo	Offline	CLO2, CLO3, CLO4	IT	Core: Using GenAI to accelerate analysis (EDA prompts, chart captions) + verification. Refs: NIST AI RMF / GenAI profile (summary), academic integrity note. Assets: prompt templates + verification checklist.	link49-1	Before class: skim materials & identify dataset/business questions. In class: follow demo; take notes. After class: complete short warm-up exercises.	
50	Module 6 – Responsible AI & Integrated Analytics M6.1 GenAI-assisted workflow (practical) (Part 2/3) • Citation & disclosure of AI assistance • Guided practice (hands-on)	Offline	CLO2, CLO3, CLO4	IT	Core: Using GenAI to accelerate analysis (EDA prompts, chart captions) + verification. Refs: NIST AI RMF / GenAI profile (summary), academic integrity note. Assets: prompt templates + verification checklist.	link50-1	In class: complete guided lab steps; record outputs. After class: finish practice tasks; update notebook/report draft.	
51	Module 6 – Responsible AI & Integrated Analytics M6.1 GenAI-assisted workflow (practical) (Part 3/3) • Citation & disclosure of AI assistance • Lab + checkpoint / reflection	Offline	CLO2, CLO3, CLO4	IT	Core: Using GenAI to accelerate analysis (EDA prompts, chart captions) + verification. Refs: NIST AI RMF / GenAI profile (summary), academic integrity note. Assets: prompt templates + verification checklist.	link51-1	In class: finish mini-lab and checkpoint. After class: submit checkpoint deliverables (code + brief answers).	
52	Module 6 – Responsible AI & Integrated Analytics M6.2 Responsible AI + Progress Test 2 (Part 1/3) • Bias, privacy, data governance, reproducibility • Prompt hygiene & logging • Concepts & instructor demo	Offline	CLO2, CLO4	IT	Core: Dashboard storytelling & communicating results; reproducibility package. Refs: Wilke – narrative & design; project packaging checklist. Assets: project report template + Git repo structure guide.	link52-1 link52-2	Before class: skim materials & identify dataset/business questions. In class: follow demo; take notes. After class: complete short warm-up exercises.	
53	Module 6 – Responsible AI & Integrated Analytics M6.2 Responsible AI + Progress Test 2 (Part 2/3) • In-class test (theory + practice) • Guided practice (hands-on)	Offline	CLO2, CLO4	IT	Core: Dashboard storytelling & communicating results; reproducibility package. Refs: Wilke – narrative & design; project packaging checklist. Assets: project report template + Git repo structure guide.	link53-1 link53-2	In class: complete guided lab steps; record outputs. After class: finish practice tasks; update notebook/report draft.	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
54	Module 6 – Responsible AI & Integrated Analytics M6.2 Responsible AI + Progress Test 2 (Part 3/3) • In-class test (theory + practice) • Lab + checkpoint / reflection	Offline	CLO2, CLO4	IT	Core: Dashboard storytelling & communicating results; reproducibility package. Refs: Wilke – narrative & design; project packaging checklist. Assets: project report template + Git repo structure guide.	link54-1 link54-2	In class: finish mini-lab and checkpoint. After class: submit checkpoint deliverables (code + brief answers).	
55	Module 6 – Integration & Assessment Clinic M6.3 Assignment clinic (Asm1–Asm3) (Part 1/3) • Rubric walk-through + exemplar review • Debugging common mistakes • Concepts & instructor demo	Offline	CLO1–CLO4	IT	Clinic materials: Asm3 rubric, common issues, exemplar submissions. Assets: office-hour checklist + Q&A board.		Before class: skim materials & identify dataset/business questions. In class: follow demo; take notes. After class: complete short warm-up exercises.	
56	Module 6 – Integration & Assessment Clinic M6.3 Assignment clinic (Asm1–Asm3) (Part 2/3) • Peer review & resubmission plan • Guided practice (hands-on)	Offline	CLO1–CLO4	IT	Clinic materials: Asm3 rubric, common issues, exemplar submissions. Assets: office-hour checklist + Q&A board.		In class: complete guided lab steps; record outputs. After class: finish practice tasks; update notebook/report draft.	
57	Module 6 – Integration & Assessment Clinic M6.3 Assignment clinic (Asm1–Asm3) (Part 3/3) • Peer review & resubmission plan • Lab + checkpoint / reflection	Offline	CLO1–CLO4	IT	Clinic materials: Asm3 rubric, common issues, exemplar submissions. Assets: office-hour checklist + Q&A board.		In class: finish mini-lab and checkpoint. After class: submit checkpoint deliverables (code + brief answers).	
58	Module 5 – Business Intelligence with Power BI M5.4 DAX practice clinic (Part 1/2) • Measures vs calculated columns • Time intelligence basics (optional) • Validate KPI logic with test cases	Offline	CLO4	IT	Project workshop pack: final rubric, presentation template, submission checklist. Assets: final project submission form.		Before class: review DAX notes and bring draft measures. In class: complete guided DAX exercises. After class: implement/validate measures on your dataset (home).	
59	Module 5 – Business Intelligence with Power BI M5.5 Dashboard UX & interaction design (Part 2/2) • Layout, hierarchy, and accessibility • Slicers, drill-through, tooltips • Avoid misleading visuals in dashboards	Offline	CLO3–CLO4	IT	Project workshop pack: final rubric, presentation template, submission checklist. Assets: final project submission form.		In class: redesign dashboard layout following UX checklist. After class: apply improvements and document design choices.	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
60	Module 5 – Business Intelligence with Power BI M5.6 Publish & governance essentials • Publishing, sharing, permissions (conceptual) • Documentation & reproducibility checklist • Final Q&A and exam preparation briefing (no project evaluation)	Offline	CLO2–CLO4	IT	Project workshop pack: final rubric, presentation template, submission checklist. Assets: final project submission form.		In class: complete publish/governance walkthrough; finalize documentation checklist. After class: self-study for practical/final exams; continue project work at home.	
61	Module 6 – PROJECT GROUP EVALUATION (EXTRA) M6.E1 Final project presentations • Evaluation ONLY (no new teaching) • Demo + Q&A; rubric scoring • Peer/individual contribution verification	Offline	CLO1–CLO4	IT	Project workshop pack: final rubric, presentation template, submission checklist. Assets: final project submission form.		In class: deliver group project presentation & demo and answer Q&A (assigned teams).	
62	Module 6 – PROJECT GROUP EVALUATION (EXTRA) M6.E1 Final project presentations • Evaluation ONLY (no new teaching) • Demo + Q&A; rubric scoring • Peer/individual contribution verification	Offline	CLO1–CLO4	IT	Project workshop pack: final rubric, presentation template, submission checklist. Assets: final project submission form.		In class: deliver group project presentation & demo and answer Q&A (assigned teams).	
63	Module 6 – PROJECT GROUP EVALUATION (EXTRA) M6.E1 Final project presentations • Evaluation ONLY (no new teaching) • Demo + Q&A; rubric scoring • Peer/individual contribution verification	Offline	CLO1–CLO4	IT	Project workshop pack: final rubric, presentation template, submission checklist. Assets: final project submission form.		In class: deliver group project presentation & demo and answer Q&A (assigned teams).	
64	Module 6 – PROJECT GROUP EVALUATION (EXTRA) M6.E1 Final project presentations • Evaluation ONLY (no new teaching) • Demo + Q&A; rubric scoring • Peer/individual contribution verification	Offline	CLO1–CLO4	IT	Project workshop pack: final rubric, presentation template, submission checklist. Assets: final project submission form.		In class: deliver group project presentation & demo and answer Q&A (assigned teams).	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
65	Module 6 – PROJECT GROUP EVALUATION (EXTRA) M6.E1 Final project presentations • Evaluation ONLY (no new teaching) • Demo + Q&A; rubric scoring • Peer/individual contribution verification	Offline	CLO1–CLO4	IT	Project workshop pack: final rubric, presentation template, submission checklist. Assets: final project submission form.		In class: deliver group project presentation & demo and answer Q&A (assigned teams).	
66	Module 6 – PROJECT GROUP EVALUATION (EXTRA) M6.E1 Final project presentations • Evaluation ONLY (no new teaching) • Demo + Q&A; rubric scoring • Peer/individual contribution verification	Offline	CLO1–CLO4	IT	Project workshop pack: final rubric, presentation template, submission checklist. Assets: final project submission form.		In class: deliver group project presentation & demo and answer Q&A (assigned teams).	

0 Constructive question(s)

6 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	N
1	Assignments (Asm1–Asm3)	on-going	3	15.0%	>0	At home	CLO1, CLO2, CLO3, CLO4	Coursework / mini-projects	Rubric-based	Asm1: Sessions 1–15; Asm2: Sessions 16–42; Asm3: Sessions 43–58	Individual submission	N
2	Group Project	on-going	1	30.0%	2	EXTRA Slots 61–66	CLO1, CLO2, CLO3, CLO4	Project presentation + demo + Q&A	Rubric-based	Self-study development across Slots 25–60; evaluation in extra slots 61–66	Lecturers; repository or structured project folder + README + presentation	In h re fo s c o V a o
3	Progress test	on-going	1	5.0%	>0	45'	CLO1, CLO2, CLO3, CLO4	Applied multiple choices / short interpretation / data reasoning	45	Integrated knowledge checkpoint	In class, by instructor	N
4	Final exam	Final exam	2	50.0%		150'	CLO1, CLO2, CLO3, CLO4			All subjects in syllabus		T > A T 5
5	Practical Exam	Final exam	1	30.0%	2	90'	CLO1, CLO2, CLO3, CLO4	Hands-on (Python/Power BI) + short interpretation/storytelling	N/A	Build clean dataset + visualization/dashboard within time limit	By exam board	In h re to a th p s

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	N
6	Theoretical exam	Final exam	1	20.0%	4	60'	CLO1, CLO2, CLO3, CLO4	Applied multiple choices + case-based interpretation / chart critique	40–60	All topics	By exam board, using computer	T q m u d le to p o